We know that $s_y = u_y t + \frac{1}{2} a t^2$ for vertical motion. Here $s_y = h$, $t = T$, $u_y = 0$ (i.e., no initial vertical velocity). Then

$$h = \frac{1}{2} g T^2 \quad \text{or} \quad T = \sqrt{\frac{2h}{g}}$$

Thus, the time of flight for projectile motion depends on the height of the tower, but is independent of the horizontal velocity of projection. If one ball falls vertically and another ball is projected horizontally with some velocity, both the balls will reach the bottom at the same time. This is illustrated in the Figure 2.40.

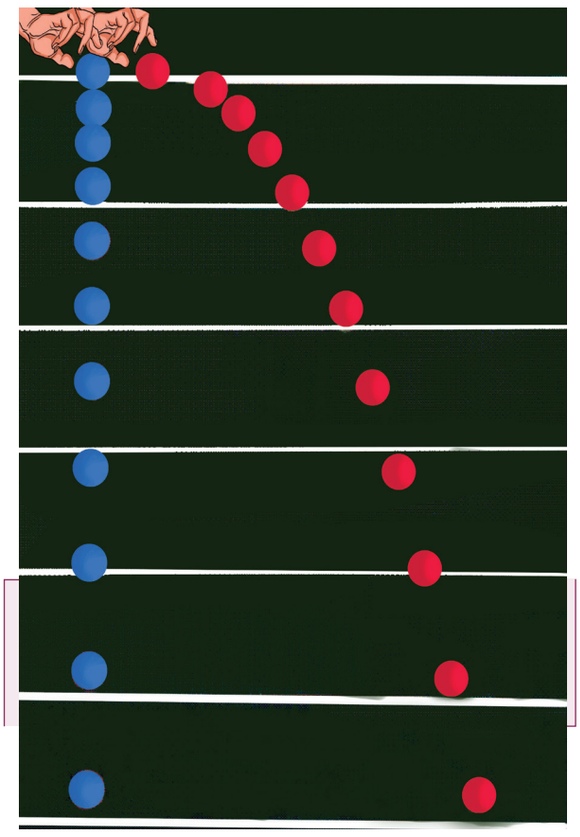


Figure 2.40 Vertical distance covered by the two particles is same in equal intervals.

(2) Horizontal range: The horizontal distance covered by the projectile from the foot of the tower to the point where the projectile hits the ground is called *horizontal range.* For horizontal motion, we have

$$s_x = u_x t + \frac{1}{2} a t^2 .$$

Here, s_x = R (range), $u_x = u$, a = 0 (no horizontal acceleration) T is time of flight. Then horizontal range = uT.

Since the time of flight $T = \sqrt{\frac{2h}{g}}$, we substitute this and we get the horizontal range of the particle as $R = u\sqrt{\frac{2h}{g}}$

The above equation implies that the range R is directly proportional to the initial velocity u and inversely proportional to acceleration due to gravity g.

(3) Resultant Velocity (Velocity of projectile at any time): At any instant t, the projectile has velocity components along both x-axis and y-axis. The resultant of these two components gives the velocity of the projectile at that instant t, as shown in Figure 2.41.

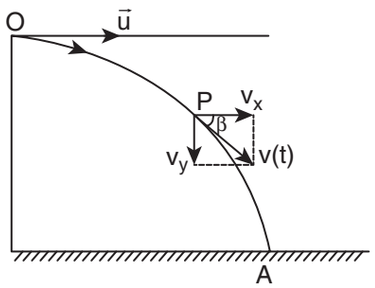


Figure 2.41. Velocity resolved into two components

The velocity component at any t along horizontal (x-axis) is $v_x = u_x + a_x t$

Since, $u_x = u$, $a_x = 0$, we get

$$v_x = u \qquad \rightarrow (2.26)$$

UNIT-2(XI-Physics_Vol-1)-1.indd 83 05-01-2022 22:15:32